

Indr 252 Applied Statistics
Department of Industrial Engineering
Koç University

Spring 2026

Syllabus

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BS&MS in IE (ODTÜ)

PhD in Operations (Case Western Reserve University)

Post-doctorate at EURANDOM in the Netherlands (A research institute on random processes)

At Koç University Industrial Engineering since 2001

Research interests: Health care applications, stochastic models

Personal interests: Books, cats, volunteering, women in STEM



Plan of the day

Syllabus

Sampling and statistics

Indr 252 Applied Statistics

Teaching assistants: Ege Oktay, Sertaç Derya, Sudem Türkoğlu, Yağız Efe İnce, Zeynep Çınar

Textbook: Statistics for Engineers and Scientists, 5th Edition (ISBN-13: 978-1260547887), W. Navidi, McGraw-Hill Education

McGraw-Hill Education Connect: Textbook companion,
<https://connect.mheducation.com/connect/container/launch.htm>

You need to buy the book and access code to be able to accomplish the reading assignments, etc.

Software to carry on the statistical analyses: Python

- Num Python: <https://numpy.org/doc/stable/reference/routines.statistics.html>
- Sci Python: <https://docs.scipy.org/doc/scipy/reference/stats.html>

All information about the course:

<https://learn.hub.ku.edu.tr/course/view.php?id=3029>

Course Description

- Visual tools to understand the data: all kinds of charts
- How to summarize data: useful statistics such average, standard deviation, etc.
- How to characterize the main properties of processes using data: parameter estimation, confidence intervals, and hypothesis testing.
- How to characterize the underlying probabilistic features using data: Distribution fitting, goodness of fit tests. Independence tests and contingency tables.
- How to relate different variables: Simple linear regression and correlation analysis. Nonlinear and multiple regression, analysis of categorical data.

Industrial engineering applications in quality control and demand forecasting. Statistical software packages and computer implementations.

Upon completion of this course, you should be able to:

Formulate a problem about a system or process as a statistical question

Design an appropriate data set and collect the data

Present the data in visual forms that help understanding potential problems

Understand and carry on the required statistical analyses to answer the chosen question

Note the shortcomings of the statistical analyses due to the underlying assumptions and data collection process

Assess statistical inferences on the underlying problem.

Learning experiences

Learning material: introducing the statistical techniques that IEs use

Experiencing:

- Letting the students experience being an IE through cases.
- Labs and case studies are designed so that students actively analyze data using appropriate software.

Tentative schedule of the course

Spring 2026	START	Tue	Thu	Lab Content Thursday
Week 1	9-Feb	Introduction	Random sampling and data description	--
Week 2	16-Feb	Random sampling and data description / Propagation of error	Propagation of error	Lab 1
Week 3	23-Feb	Point estimation	Point estimation	Lab 2 - Generating random variables-descstats-plots
Week 4	2-Mar	Probability plots/CLT	Interval estimators	Lab 3 - Analyzing Dataset
Week 5	9-Mar	Interval estimators	Interval estimators	Lab 4 - CI
Week 6	16-Mar	Spring Break		
Week 7	23-Mar	Interval estimators	Interval estimators	Sample Case
Week 8	30-Mar	Tests of hypotheses	Case Exam 1: April 2 16:00-18:30	
Week 9	6-Apr	Tests of hypotheses	Tests of hypotheses	Lab 5 - HT
Week 10	13-Apr	Tests of hypotheses	Tests of hypotheses	PS
Week 11	20-Apr	Linear regression	4/23/2026	
Week 12	27-Apr	Linear regression	Linear regression	MIDTERM: EXACT DATE - TBA
Week 13	4-May	Linear regression	Multiple regression	Lab 6 - ChiSq Distribution Tests
Week 14	11-May	Multiple regression	Multiple regression	Lab 7 - Multiple Regression
Week 15	18-May	5/19/2026	Case Exam 2: May 21 16:00-18:30	

Assessment Methods

Method	Description	Weight %
Case Study 1	Case Exam	10.00
Case Study 2	Case Exam	15.00
Midterm	Exam	25.00
Final Exam	Comprehensive Final exam	30.00
Reading Assignments	Assignments from the book	5.00
Participation	Participation (in-class exercises)	5.00
Participation	Lab work (Lab assignments)	10.00
Case Study 1	Case Exam	10.00
Case Study 2	Case Exam	15.00

Lectures

Lectures: In-class participation: class exercises (5%)

Students will be given straightforward questions in the lectures as class exercises. Students can work on these exercises in groups of two (never more than two). Note that putting in an honest effort to solve the assignment will result in full credit.

Labs (10%)

Thursdays at 17:30-18:40 in SNAB149

- Aim: to teach how to carry out statistical computations using Python and how to interpret the computations and plots.
- There are seven lab sessions in which Python functions are used to perform specific statistical tasks. Labs 2-7 require students to complete an assignment.
- Lab assignments require working on two main tasks: (1) preparing the Python code that computes all the necessary statistics and generates the plots. (2) interpreting these statistics and plots. The assignment guides students in exploring the meaning of the statistics and plots.

Labs (10%)

Thursdays at 17:30-18:40 in SNAB149

The lab assignments will be uploaded to LearnHub on the day of the lab session.

Students will have access to the assignment description, a preliminary Python script, and a video describing the lab. Students will have 30 minutes to work on their Python code. The teaching team will help the students who have difficulty with the Python code. Then we will discuss their findings for 15 minutes. In the rest of the lab session, the students will write their reports and submit their work at the end.

The Python code given to the students is usually sufficient to complete the assignment. Still, students may use AI to generate parts of the Python code. However, AI can only be used to correct the English in the report.

One lab session is dedicated to a sample case study to prepare you for the case exams. You will be given a short test on LearnHub at the end of this session. In two lab sessions, we will hold case exams.

PSs & Sample case

- Sample case: all students should read and think about the case before the session; during the sessions, random groups will work on one part of the case, and then we will discuss the results altogether. You will be given a short test on LearnHub at the end of this session. (as part of lab assessments)
- Problem sessions: problems on the course topics will be solved by the TAs

Reading assignments (5%):

The corresponding book chapters are assigned for reading before/after the lectures, so that you can be prepared for the new subjects and digest the material covered in class.

Once you read the related chapters, you will be presented a number of questions on the basic concepts. We will hold Q&A sessions in the beginning of each lecture on the latest reading assignment to discuss your questions and comments on these concepts.

These assignments will help you to assess your level of understanding in the course.

Exams

One midterm (25%): multiple-choice questions in computer labs.

Two case exams (10% & 15%): Group exam; outputs are a Python file and a report

Final exam (25%): A comprehensive exam with multiple-choice questions in computer labs.

Make-up Exam Policies: A make-up exam for the case exam and a make-up exam for other tests will be given to those who have a valid excuse at the end of the semester. Both make-up exams will be comprehensive.

Case Exams

- On two Thursdays through the lecture and lab sessions (16:00-18:30).
 - in groups in the computer labs, where physical presence is mandatory. Students can form their groups. Students without a group will be assigned to a group by the teaching team.
 - Students will not have access to cell phones, calculators, or the internet during the exams.
 - In these exams, students are supposed to analyze a case statistically, which will require using Python code. Note that students are not allowed to access any AI tools; instead, some partial code will be shared with them during the exam. Hence, students need to learn to write or modify Python code to meet their specific needs.

LearnHub and Connect

I will share the information how to register Connect and access the e-book on Wednesday.

Reading assignments:

- Aim to help you in following the course.
- Some midterm questions will be similar to these exercises.

If you cannot afford buying the book, please contact me.

Letter grading: Catalogue

Letter Grade	Scale	Grade Range
A+	4	95-100
A	4	90 – 94
A-	3.7	85 – 89
B+	3.3	80 – 84
B	3	75 – 79
B-	2.7	70 – 74
C+	2.3	65 – 69
C	2	60 – 64
C-	1.7	56 – 59
D+	1.3	53 – 55
D	1	50 – 52
F	0	0 – 49

Academic Dishonesty

Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct (<http://vpaa.ku.edu.tr/academic/student-code-of-conduct>). Actions considered as academic dishonesty at Koç University include but are not limited to cheating, plagiarism and collusion. Copying from others or providing answers or information, written or oral, to others is cheating. Copying from another student's paper or from another text without written acknowledgment is plagiarism. Unauthorized help from another person or having someone else write one's paper or assignment is collusion.

Cheating, plagiarism and collusion are serious offenses resulting in an F grade and disciplinary action.

Other issues

Grades: To be announced in LearnHub. All objections to grades should be made within 7 days of their announcement.

Why Statistics Matters in Industrial Engineering?

“Without data, you’re just another person with an opinion.” — W. Edwards Deming

How many of your daily or engineering decisions are truly based on data?



Everyday Variation

Why Did My Flight Get Delayed?

- Which airline is more reliable?
- Does the average delay tell the full story?

→ Statistics helps us understand variation, not just averages.

Decision Under Uncertainty

Should We Buy a New Machine?

Current defect rate: 2%

New machine: 1.5%

Sample: 200 parts → 3 defects

Would you invest 1 million TL based on that?

→ Statistics quantifies confidence.

Statistical Thinking

Statistical inference = separating signal from noise.

This is the backbone of data-driven industrial decisions.

Amazon's A/B Test

Amazon changes button color → Sales ↑ 3%.

Is it real or random?

Experiments = foundation of Design of Experiments (DOE).

Industrial Experiments

Examples:

- Testing a new assembly method
- Trying a new supplier material
- Changing oven temperature

Same logic = process optimization.

The Curious Case of the Broken Bolt

Measurements all within spec, but drifting upward.

Would you stop production?

Process Stability

Control charts show process health.

Predictable = good.

Statistics = early warning system.

Recap

Flights → Variation

Machine → Decision under uncertainty

Amazon → Experiments

Bolts → Process monitoring

What You'll Learn

Descriptive statistics

Probability & sampling

Inference & regression

Design of experiments

Statistical quality control

Real-World Impact

Manufacturing → Process improvement

Logistics → Forecasting demand

Tech → A/B testing

Industrial engineers = translators between data and decisions.

Closing Thought

“Statistics is not about numbers — it’s about learning from data.”